DSA Assessment — Test 2

Data Structures & Algorithms — Searching & Sorting

Name: MUSKAN SAHANI	Date: ________
Time: 1 Hour	Total Marks: 30

Instructions:

• Section A: Dry Run — Fill in the tables below or in a txt file

• Section B: Code — Create .java files in VS Code, run and verify output

• Attempt all questions — Show every step clearly for full marks

SECTION A — Dry Run (15 Marks)

Q1. Linear Search Trace (5 Marks)

```
Array = {14, 52, 7, 38, 21, 63, 45} Target = 63
```

Trace the search step by step:

Step	Index	Value Checked	Match? (Yes/No)
1	0	14	No
2	1	52	No
3	2	7	No
4	3	38	No
5	4	21	No
6	5	63	Yes

Q1a. Found at index = 5

Q1b. Total steps taken = 6

Q1c. Is this Best Case, Worst Case or Average Case? Explain in one line:

It was near to the worst case because we need to traverse the array till the second last element.

Q2. Binary Search Trace (5 Marks)

```
Array = {6, 14, 23, 39, 47, 55, 72} Target = 23
```

Complete all steps using formula: Mid = (Left + Right) / 2

Step	Left	Right	Mid	Value at Mid	Decision
1	0	6	3	39	Not found
2	0	2	1	14	Not found
3	2	2	2	23	Found

Q2a. Found at index = 2

Q2b. Total steps for Binary Search = 3

Q2c. Total steps if Linear Search was used instead = 3

Q2d. Which is faster and why? (one line) Binary Search because of time complexity

Q3. Bubble Sort Trace (5 Marks)

```
Array = {38, 15, 42, 6, 29}
```

Pass 1: (Compare adjacent elements, swap if left > right)

Step	Elements Compared	Swap? (Yes/No)	Array After Step
1	38 and 15	yes	15,38,42,6,29
2	38 and 42	no	15,38,42,6,29
3	42 and 6	yes	15,38,6,42,29
4	42 and 29	yes	15,38,6,29,42

Which element settled at the END after Pass 1? = 42

Pass 2: (Sorted elements at end are ignored)

Step	Elements Compared	Swap? (Yes/No)	Array After Step
1	15 and 38	no	15,38,6,29,42
2	38 and 6	yes	15,6,38,29,42
3	38 and 29	yes	15,6,29.38.42

Which element settled after Pass 2? = 38

Bonus Thinking Question (No marks — just think!):

If the array was already sorted {6, 15, 29, 38, 42} — how many swaps would Bubble Sort do? Why?

No swaps, because its already sorted

SECTION B — Code (15 Marks)

Create separate .java files for each question. Run and verify your output matches expected.

Q4. Linear Search Code (5 Marks)

File name: MyLinearSearch.java

```
Array = {33, 8, 71, 19, 56, 44, 27} Target = 56
```

Your code must:

- Print — Checking index 0 value 33 — for every step
- Print Found or Not Found at the end
- If found — print the index number
- Print total number of steps taken

Expected Output:

```
Checking index 0 value 33
Checking index 1 value 8
Checking index 2 value 71
Checking index 3 value 19
Checking index 4 value 56
Found at index 4
Total Steps = 5
```

Q5. Binary Search Code (5 Marks)

File name: MyBinarySearch.java

```
Array = {9, 21, 34, 48, 57, 66, 83} Target = 66
```

Your code must:

- Print Left, Right, Mid and Value at every step
- Print Found or Not Found

■ Print index if found

■ Print total steps taken

Expected Output:

```
Left=0 Right=6 Mid=3 Value=48
Left=4 Right=6 Mid=5 Value=66
Found at index 5
Total Steps = 2
```

Q6. Selection Sort Code (5 Marks)

File name: MySelectionSort.java

```
Array = {52, 18, 37, 9, 64}
```

Your code must:

■ Print array before sorting

■ Print assumed minimum at start of every pass

■ Print when new minimum is found

■ Print array after every pass

■ Print total swaps at the end

■ Print final sorted array

Expected Output:

```
Before Sorting: 52 18 37 9 64
--- Pass 1 ---
Assumed minimum: 52 at index 0
New minimum found: 18 at index 1
New minimum found: 9 at index 3
Swapped: 9 to index 0
Array after Pass 1: 9 18 37 52 64
... and so on ...
Total Swaps: 2
After Sorting: 9 18 37 52 64
```

Marking Scheme

Section	Question	Marks
Section A	Q1 Linear Search Trace	5
	Q2 Binary Search Trace	5
	Q3 Bubble Sort Trace	5
Section B	Q4 Linear Search Code	5
	Q5 Binary Search Code	5
	Q6 Selection Sort Code	5
	TOTAL	**30**

Best of Luck! ■ — Read every question carefully before answering!